

Table 1 Differences in diagnoses, gender and family status for participants with a suicide attempt and those without a suicide attempt within the 12-month follow-up interval

	Participants with no suicide attempt (<i>n</i> = 132) ^a		Participants with a suicide attempt (<i>n</i> = 43) ^b		χ^2	d.f.	<i>P</i>
	<i>n</i>	%	<i>n</i>	%			
ICD-10 diagnoses							
F0	1	0.76	0	0.00	0.00	1	1.00
F1	17	12.88	12	27.91	4.39	1	0.04
F2	1	0.76	0	0.00	0.00	1	1.00
F3	106	80.30	31	72.09	0.74	1	0.39
F4	42	31.82	17	39.53	0.61	1	0.43
F5	5	3.79	5	11.63	2.44	1	0.12
F6	20	15.15	19	44.19	14.48	1	0.00
F7	0	0.00	0	0.00	–	–	–
F8	1	0.76	0	0.00	0.00	1	1.00
F9	2	1.52	1	2.33	0.00	1	1.00
Gender					3.09	2	0.21
Female	75	56.8	24	55.8			
Male	57	43.2	18	41.9			
Diverse	0	0	1	2.3			
Family status					4.87	4	0.30
Single	55	41.7	14	32.6			
Partnership	25	18.9	9	20.9			
Married	27	20.5	5	11.6			
Divorced	20	15.2	11	25.6			
Widowed	1	0.8	1	2.3			

F0: Organic, including symptomatic, mental disorders; F1: Mental and behavioural disorders due to psychoactive substance use; F2: Schizophrenia, schizotypal and delusional disorders; F3: affective disorders; F4: Neurotic, stress-related and somatoform disorders; F5: Behavioural syndromes associated with physiological disturbances and physical factors; F6: Disorders of adult personality and behaviour; F7: Mental retardation; F8: Disorders of psychological development; F9: Behavioural and emotional disorders with onset usually occurring in childhood and adolescence.

a. 75.43% of the total sample with full information on suicide reattempts within the entire 12-month follow-up interval.

b. 24.57% of the total sample with full information on suicide reattempts within the entire 12-month follow-up interval.

Table 2 shows that participants with a suicide attempt within the 12-month follow-up interval had significantly higher baseline scores for depression, perceived burdensomeness, thwarted belongingness, hopelessness, suicidal ideation and number of lifetime suicide attempts. However, out of the four indicators applied to measure capability for suicide (GCSQ total score, the fearlessness about death and pain tolerance subscales, and the perceived capability for suicide item), only the perceived capability item differed significantly between the two groups, with higher mean scores in the attempter group than in the non-attempter group.

Moreover, a significantly larger fraction of the participants with a suicide attempt within the 12-month follow-up interval had ICD-10 F1 or F6 diagnoses (Table 1).

Multifactorial prediction of suicide attempts

Results of a hierarchical linear regression analysis entering perceived burdensomeness, thwarted belongingness and capability for suicide in the first step, two-way interactions in the second step and the three-way interaction in the third step are summarised in Table 3. Perceived burdensomeness showed a significant main effect on suicide attempts within the 12-month interval, which remained significant even after entering all interaction terms into the model. No other predictor was significantly related to the dependent variable. Perceived burdensomeness at baseline was entered into a ROC analysis. Results revealed a moderate performance of perceived burdensomeness in screening future suicide attempt risk (AUC = 0.729; 95% CI 0.645–0.814; $P < 0.01$).

Discussion

The current study examined a central assumption of the IPTS regarding the prediction of future suicide attempts in a high-risk

sample of adults admitted to a psychiatric hospital because of a suicide attempt or severe suicidal ideation within 12 months after the initial assessment. The primary finding was that, contrary to expectation, the interaction between thwarted belongingness, perceived burdensomeness and capability for suicide did not predict suicide attempt in the 12-month follow-up-period. Although unexpected and contrary to the theoretical assumptions, this finding is consistent with retrospective studies in in-patient populations^{40,41} as well as two prospective studies in adolescent populations,^{11,18} neither of which found support for this main assumption of the IPTS. To our knowledge, Joiner et al¹⁰ conducted the only study that found the three-way interaction to be predictive of suicide attempts in an in-patient sample. However, that study – and also the prospective studies by Czyz et al¹⁸ and King et al¹¹ – made use of proxy measures to assess capability for suicide and focused on past instead of future suicide attempts. Unlike all previous studies, the current study made use both of validated measures originally developed to assess the theory's main constructs and a prospective study design.

Overall, the results of the current study question the relative importance and utility of the three IPTS constructs in explaining suicidal behaviour – at least with regard to in-patient samples. However, from a clinical perspective, knowledge about risk factors is especially important in dealing with high-risk patients. Therefore, the results challenge the clinical utility of employing the IPTS to predict suicide risk (see⁷). Still, in line with previous research,^{7,9} the current results highlight the importance of perceived burdensomeness in understanding suicidal ideation and behaviour. Perceived burdensomeness showed a significant main effect on suicide attempt within the 12-month interval, which remained significant even after entering all interaction terms into the regression model. However, the direct pathway between perceived burdensomeness and suicide attempts is not part of the IPTS. According to the IPTS, one would expect a main effect of perceived